

Kryštof Przeczek

(+44) 7575 403039 | kprzeczek4@gmail.com | <https://kryzp.github.io> | <https://github.com/kryzp>

PROFILE

I am a Computer Science student focused on graphics programming, real-time rendering, and simulation. I have experience building Vulkan renderers, path tracers, and physics simulations from scratch in C/C++. I have an interest in performance-critical systems combining computer science, mathematics, physics, and visual art.

SKILLS

- **Languages:** C, C++, C#, GLSL, HLSL, Slang (rasterisation and compute tasks), Java, and Python.
- **Graphics APIs:** OpenGL and Vulkan.
- **Tools:** Git, Unix terminal, RenderDoc, Xcode, and Radeon Developer Suite.
- **Engines & Frameworks:** C# FNA/MonoGame library, Unity, Godot, and GameMaker Studio 2.

PROJECTS

Magpie | *C, SDL, Vulkan, ImGui* <https://github.com/kryzp/magpie>
Vulkan renderer with hot-reloadable code and assets, async asset streaming, a fiber-based job system, physically-based materials, with a render graph and scene system for efficient GPU synchronization and geometry management.

Stick it Together | *FastAPI, Kotlin, React, ElevenLabs* <https://github.com/kryzp/sotonhack26backend>
Winner of SotonHack '26. 1v1 party game linking concepts via semantic similarity scoring. Built with Kotlin, React, and FastAPI; uses Gemini for scoring and ElevenLabs for real-time voice commentary.

Path Tracer | *C++* <https://github.com/kryzp/pathtracer>
Path tracer capable of physically-based metallic, lambertian, and refractive materials.

2D Finite-Element Solver | *Java* <https://github.com/kryzp/simple-finite-element-solver>
Computes beam deflection via stiffness matrices built from a 2D structure and boundary conditions.

2D Navier-Stokes Fluid Solver | *C++* <https://github.com/kryzp/simple-navier-fluid-sim>
Fluid simulation using iterative projection methods to solve for diffusion, advection and external forces.

WORK EXPERIENCE

Microsoft June 2022
Summer Work Placement

- Shadowed a senior software engineer to learn about the tools and algorithms used daily to ensure rapid response times in their high-speed communications network.

EDUCATION

University of Southampton 2024–Present
Computer Science MEng

Hills Road Sixth Form College, Cambridge 2022–2024
Further Maths A, Maths A, Computer Science A, Physics B

ACHIEVEMENTS

- **Programming Competitions:** Competed in Advent of Code (2022–Present), placing 1st in the school (out of 150) in Year 13, and 2nd in Year 12.
- **Olympiads:** Earned Silver in both the British Informatics Olympiad (BIO) and British Physics Olympiad (BPhO).
- **Project Euler:** Completed 104 problems independently, ranking me in the top 5% of all competitors.
- **Duke of Edinburgh's Award (DofE):** Refined skills in game development, swimming, and volunteering at a local library.

HOBBIES

- Analysing games by intercepting CPU-to-GPU data to study their graphics techniques.
- Experimenting with esoteric shaders on Shadertoy to create visually striking scenes.
- Sketching landscapes and painting wildlife digitally.
- Playing guitar, I'm a fan of post-rock and post-apocalyptic styles.
- Undertaking more hands-on projects like building a crossbow and carpentry.